

DataX — Outlet Intelligence Final Round Submission

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Executive summary

The problem statement specifies that historical monthly volume $V_{\{i,t\}}$ is the censored minimum of latent demand $D_{\{i,t\}}$ and a binding systemic constraint $C_{\{i,t\}}$ (credit, stockout, no cooler, delivery cap):

$$V_{\{i,t\}} = \min(D_{\{i,t\}}, C_{\{i,t\}})$$

There is no observed target. Our framework estimates D for each of 20,000 outlets for January 2026 using a five-method statistical ensemble anchored on peer-conditional quantiles, validates the result against held-out January 2025, allocates a fixed LKR 5,000,000 trade-spend budget over 8,588 Western Province outlets through a concave water-filling optimiser, and surfaces every prediction with per-outlet feature attribution and an LLM-generated business narrative inside a Next.js web application.

Key headline numbers:

Metric	Value
Outlets predicted for January 2026	20,000
Engineered features per outlet	229
Statistical methods cross-checked	5
Pairwise method Spearman ρ	all ≥ 0.83
Hold-out Spearman ρ (Jan 2025 predicted from 2023-24 only)	0.793 over 10,949 outlets
LKR 5M actually allocated	4,999,912.87 across 8,416 funded outlets
Channel split (Discount / Merch / Promo)	34.9% / 41.8% / 23.3%
Cooler Top-100 24-month margin uplift	LKR 36,650,325 (net 31.65M of 5M capex)
Dormancy classifier 5-fold CV AUC	0.879 $\pm$ 0.004
HDBSCAN sub-province sales territories	91

a. Data Engineering and Scraping Pipeline

Lakehouse architecture

The codebase is organised as a Bronze $\rightarrow$ Silver $\rightarrow$ Gold lakehouse, orchestrated by a 12-phase entry point at `src/run_pipeline.py`. Bronze copies the five raw CSVs (plus the Excel codebook) byte-for-byte into `data/bronze/`, computes a sha256 manifest, and asserts cross-file `Outlet_ID` set equality (0 orphans, 0 ghosts in this dataset). Silver applies the data-quality library to produce cleaned parquets in `data/silver/clean/` and persists every rejected row with a documented `_rejection_reason` to `data/silver/quarantine/`. Gold engineers the 229-column per-outlet feature frame plus the dense (Outlet $\times$ Month) panel of 720k rows.

Reusable data-quality library

src/dq_checks.py exposes six parameterised functions:

Function	Purpose
check_duplicates(df, primary_keys)	Both rows of a duplicate go to rejected (keep nothing)
check_nulls(df, mandatory_columns)	Rejects rows where any mandatory column is null
check_referential_integrity(df, fk, valid_values)	Rejects rows whose FK is not in the known set
check_value_range(df, column, min_val, max_val)	Numeric / date range validator
check_format(df, column, validator)	Regex or callable validator on a column
check_value_set(df, column, allowed_values)	Whitelist validator (case-tolerant where needed)

Every check is invoked through an `apply_check()` wrapper that persists the rejected rows and appends a summary row to `_quarantine_summary.csv`. The same six functions are used across all five datasets so the quality assurance is parameter-driven, not ad-hoc per file. Total rejected: 73,285 rows (3.08%), distributed across the five datasets.

POI acquisition (external Gold enrichment)

OpenStreetMap is queried via the Overpass API with one country-bounding-box request per tag, 36 tag combinations in total. Responses are persisted to `data/bronze/poi_raw/*.json` so subsequent runs are network-free. Retry-with-backoff handles the occasional HTTP 429. Total POIs scraped: 43,023.

Distance-decay feature engineering

The 36 POI tags are grouped into 9 semantic categories and translated into per-outlet scores using non-linear distance kernels rather than flat counts:

Category	Kernel	σ (Gaussian)	Search radius
Footfall (bus stop, fuel, bank, taxi, station)	Gaussian	150 m	500 m
School (school, university, college, kindergarten)	Gaussian	400 m	1 km
Tourist (hotel, attraction, beach, viewpoint)	Gaussian	1000 m	2 km
Health (hospital, clinic, pharmacy)	Gaussian	400 m	1 km
Worship (place_of_worship, temple)	Gaussian	300 m	1 km
Food service (restaurant, cafe, fast_food, ice_cream)	Gaussian	150 m	300 m
Leisure & rec (park, fitness, stadium)	Gaussian	500 m	1.5 km
Population (landuse=residential)	Gaussian	600 m	1.5 km
Competitor POI (supermarket, convenience, alcohol)	Gravity	—	800 m

A composite `spatial_demand_score` aggregates the nine normalised category scores with category-specific weights derived from FMCG-domain reasoning (food service 3.0, footfall 3.0, population 2.5, tourist 2.5, ...).

Non-linear competition features

Beyond raw counts (`competitors_500m` / `1km` / `2km` and same-type subsets), the team added three non-linear market-structure measures (in `src/competition_features.py`):

- **HHI (Herfindahl-Hirschman Index)** over a 1.5 km neighbourhood, using `monthly_volume_mean` as the share proxy. Reported on the standard 0-10,000 scale. Western HHI $\approx 1,101$ (fragmented), North-Western $\approx 4,319$ (concentrated).
- **Voronoi-style territory radius**: mean distance to the five nearest neighbour outlets, a proxy for how much "ground" an outlet effectively owns.
- **Type-weighted competition pressure**: sum over nearby competitors of a type-similarity weight ($\text{Grocery} \times \text{Grocery} = 1.0$, $\text{Grocery} \times \text{Pharmacy} = 0.3$, ...), so a clustered same-type market hurts more than a mixed-type one.

Replenishment-friction signal

`replenishment_friction = stockout_flag_sum / active_months` (clipped to 0-1) is the team's representation of the cooler-replenishment-cycle constraint named in the brief. High values mean the delivery / refill cycle is failing the outlet even when nominal cooler capacity exists. Top-quartile outlets are flagged with `high_friction_flag`. Network mean ≈ 0.198 , $\approx 5,050$ outlets in the high-friction band.

Feature inventory (229 columns)

The full feature frame at `data/gold/outlet_features.parquet` covers: volume statistics, activity & censoring signals, January-specific stats, master attributes, SKU mix (mass / mid / premium / super_premium share), competitor density, 9-category POI scores plus flat counts, composite spatial signal, market structure (HHI / territory / type pressure), province climate, geographic coordinates, replenishment friction. Every column is consumed downstream by the modelling, budget LP, or business-decision modules.

b. Data Cleaning

Initial assessment

A first-pass exploratory survey of the five raw CSVs surfaced eleven concrete issues. The team's policy is to **never silently drop a row**: every rejection writes to a quarantine parquet with a documented reason, and every silent correction emits a forensics finding with count + treatment.

Per-dataset quality checks

Dataset	Checks applied	Rejected (rows)
<code>outlet_master.csv</code>	PK duplicates, mandatory nulls, value-set on <code>Outlet_Size</code> / <code>Outlet_Type</code> , integer coercion on <code>Cooler_Count</code>	196 (nulls)
<code>outlet_coordinates.csv</code>	PK duplicates, mandatory nulls, referential to master, lat/lon range against Sri Lanka	236

Dataset	Checks applied	Rejected (rows)
	bounding box (5.9-9.9 °N, 79.5-82.0 °E)	
distributor_seasonality_details.csv	PK duplicates, mandatory nulls, referential to the known 10 distributors, value-set on Seasonality_Index, completeness (10 × 3 × 12 = 360)	0
holiday_list.csv	PK on (Date, Holiday_Type), mandatory nulls, ISO 8601 format, value-set on Holiday_Type	198
transactions_history_final.csv	PK on (Outlet_ID, Year, Month, SKU_ID), mandatory nulls, referential to outlet master and distributor whitelist, value-range on year / month / volume / bill	72,655

Total rejected: 73,285 rows (3.08% of the 2.38 M transaction set).

Forensic findings (beyond DQ)

Finding	Count	Treatment
Outlet_Type typos normalised (Bakry→Bakery, Grocry→Grocery, " Eatery"→Eatery, etc.)	985	cleaned silently with a logged finding
Outlet_Size lowercase "small" detected	600	flagged; 225 reclassified via cooler count, 375 case-normalised
Outlet_Size reclassified by cooler count (size = "small" but coolers ≥ 2)	225	cleaned (Medium / Large / Extra Large by Cooler_Count)
Outlet_Size imputed from cooler count (null Outlet_Size)	196	cleaned (imputed from cooler count)
Lat/Lon swapped values recovered (200 outlets had Latitude and Longitude transposed)	200	cleaned (deterministic swap when lat ∈ lon-range AND lon ∈ lat-range)
Transaction value classification (negative volume + negative bill = return; negative + positive = data error)	0 in this dataset	rule applied
Codebook references absent column (Product_Name)	1	flagged
Codebook filename divergence	1	flagged
Automated-entry signatures (≥ 6 identical consecutive Volume_Liters per outlet-SKU)	0 found	scanned, no findings
Stockout-sandwich months (zero flanked by positive months)	80,611	flagged for modelling — used as a constraint signal, not quarantined
Dead-then-resurrected outlets (≥ 6 consecutive zero months and then resumed)	2,707	flagged — used as positive training labels for the dormancy classifier
Coordinates outside coarse land-mask (≥ 5 km from any land outlet)	521	quarantined — likely GPS errors; coastal coordinates kept within 5

Finding	Count	Treatment
		km buffer

The Lat/Lon swap recovery and the size-imputation logic are deliberate: a naive bounding-box rule would have quarantined 240 useful outlets; an `is_null(Outlet_Size)` rule would have lost 196 outlets whose cooler count plainly signals their size class.

c. Mathematical Framework

The censoring structure and a note on terminology

The brief refers to the demand curve as "left-censored". In standard econometric notation $V = \min(D, C)$ is **right-censored from above**: when the constraint binds we observe C , while the true latent demand D is at least C . The brief is using the term informally; our Tobit Type I implementation treats the binding cases as right-censored exactly as Tobin (1958) specifies. The mathematics below uses the standard convention.

Cohort design

For each outlet we identify a cohort that shares (Outlet_Type, Outlet_Size, Province, POI density tier). The tier is computed within each province (Q1/Q2/Q3/Q4 of `poi_total_2km`) so an outlet's tier means the same thing across the four provinces — mitigating OSM rural coverage bias. The cohort decision walks five fallback levels until a cohort with $n \geq 30$ unconstrained outlet-months is found:

```
L0  Type × Size × Province × poi_tier    (finest)
L1  drop poi_tier
L2  drop province
L3  drop size
L4  global                                (coarsest)
```

In the current dataset > 99% of outlets resolve at L0.

Constraint detection (four OR-combined rules)

An outlet-month is flagged constrained if any rule fires:

1. **Stockout sandwich** — zero volume between two positive months
2. **Zero-in-active outlet** — $V = 0$ while the outlet has any positive month overall (intermittent supply)
3. **No-cooler + high-zero share** — Cooler_Count = 0 AND zero_months_share > 0.30 AND $V = 0$
4. **No-cooler + low-volume** — Cooler_Count = 0 AND $V > 0$ AND $V < \text{own Q90} / 3$ (suppressed cold-storage outlet operating below capacity)

About 43.6 % of outlet-months are flagged in the current data. Rule 4 was added in a second pass after the team observed that 35 % of outlets have zero coolers and many of those are clearly suppressed rather than fully inactive.

Five-method ensemble

The base prediction is the peer-conditional Q90 anchor:

```
| base_potential_i = max( own Q90 of unconstrained months_i, peer cohort Q90_i )
```

The January 2026 projection multiplies the base by three calibrated factors:

$$\text{proj}_i = \text{base}_i \times \text{seasonality_mult}(\text{Jan 2026 category, distributor}) \times \text{YoY_growth_mult}(\text{distributor}) \times \text{holiday_count_mult}$$

Seasonality multipliers are calibrated empirically per distributor and Seasonality_Index level (Favorable $\approx$ 1.30-1.43, Moderate $\approx$ 0.90-0.99, Un-Favorable $\approx$ 0.59-0.65). YoY is the geometric mean of Jan-over-Jan growth per distributor, clipped to $[0.85, 1.30]$. The holiday multiplier compares the count of unique Jan 2026 holiday dates against the historical Jan average (the inputs use `nunique(Date)` on both sides to avoid asymmetric counting). Sanity bounds: floor = own Q95, ceiling = $5 \times$ peer cohort Q99.

The peer anchor is cross-checked by four independent estimators:

- **Phase-3 business formula** (auditable):

$$\text{formula}_i = \text{Q95}(\text{own history})_i \times \text{type_mult} \times \text{size_mult} \times \text{seasonality} \times \text{holiday} \times (1 + 0.30 \text{ spatial_signal}_i) \times 1 / (1 + 0.25 \text{ competitor_density_norm}_i)$$

- **Log-linear regression** on the unconstrained subset: $\log(1+V) \sim X \beta$, with X covering attributes, SKU mix, POI scores, and one-hot categoricals.
- **Tobit Type I MLE** (Tobin 1958): full likelihood with right-censoring at the observed value when constrained, fitted by L-BFGS-B on a deterministic 300k-row sample.
- **Stochastic Frontier Analysis** (Aigner, Lovell & Schmidt 1977 half-normal): $\log V = X \beta + v - u$ with $v \sim N(0, \sigma_v^2)$, $u \sim |N(0, \sigma_u^2)|$. The frontier $\exp(X \beta)$ is the unconstrained potential; u captures the inefficiency.

Pairwise Spearman ρ in this dataset:

Pair	ρ
peer-Q90 vs log-linear	0.86
peer-Q90 vs Tobit	0.84
peer-Q90 vs SFA	0.85
peer-Q90 vs Phase-3	0.84
log-linear vs Tobit	0.92

All five methods agree on outlet ranking ($\rho \geq 0.83$ pairwise), so the five-way ensemble fires:

$$\text{final}_i = 0.40 \text{ peer}_i + 0.18 \text{ log_linear}_i + 0.18 \text{ tobit}_i + 0.16 \text{ sfa}_i + 0.08 \text{ phase3}_i$$

Peer-Q90 receives the dominant anchor weight because it is the most stable estimator on hold-out; the parametric methods share the remaining 0.60 weight with smaller mass on the formula because it is the least independent of peer-Q90 by construction.

Validation in the absence of `y`

Because there is no observed January 2026 ground truth, the team designed three internal checks plus a hold-out:

- **Internal consistency** — quantile sensitivity (Q85 / Q90 / Q95 overlap), constraint-rate plausibility (target 15-40%; observed 43.6%), magnitude ratio (predicted / observed mean), spatial Spearman of prediction against `poi_total_2km`.
- **Hold-out validation** — re-fit the entire framework on 2023 + 2024 only, predict January 2025, compare to the actual January 2025 unconstrained values. Spearman $\rho = \mathbf{0.793}$ over 10,949 outlets. The MAE and median % error are also reported in the audit CSV.

- **XAI surrogate fidelity** — an XGBoost regressor trained to reproduce the ensemble achieves Spearman $\rho \approx 0.97$ and $R^2 \approx 0.998$ against the ensemble. This is the substrate for SHAP and the counterfactual deltas.

d. Spend Optimization Logic

Problem formulation

The brief allocates a fixed LKR 5,000,000 promotional budget over Western Province for January 2026, to be split across discounts, merchandising, and promotional spend, maximising additional volume relative to historical patterns.

Let x_i be the LKR allocated to outlet i (Western Province only, 8,588 outlets after sea-coordinate quarantine) and κ_i be a per-outlet response coefficient. Promotional response is well documented in marketing-mix-modelling literature (Hanssens, Parsons & Schultz 2001; Jin et al. 2017) as a concave function of spend. We adopt the standard square-root utility shape:

$$\begin{aligned} & \max \sum_i \kappa_i \sqrt{x_i} \\ & \text{s.t. } \sum_i x_i \leq 5,000,000 \text{ LKR} \\ & 0 \leq x_i \leq 5,000 \text{ LKR} \quad (\text{per-outlet cap to prevent concentration}) \\ & i \in \text{Western Province} \end{aligned}$$

The Lagrangian admits a closed-form optimum:

$$\partial L / \partial x_i = \kappa_i / (2 \sqrt{x_i}) - \lambda = 0 \Rightarrow x_i^* = (\kappa_i / 2\lambda)^2$$

We bisect on λ until the budget constraint binds and clip each x_i^* against the per-outlet cap inside every bisection step (a clean water-filling pattern). A final flooring + scale-down step absorbs any floating-point overshoot.

Response coefficient design

$$\begin{aligned} \kappa_i &= \text{uplift}_i \times \text{cooler_priority}_i \times (1 + 0.4 \times \text{spatial_demand_score}_i) / (1 + 0.3 \times \text{competitor_density_norm}_i) \\ \text{where } \text{uplift}_i &= \max(0, \text{predicted_jan2026}_i - \text{observed_recent_avg}_i) \\ \text{cooler_priority}_i &= 1.5 \text{ if Cooler_Count} = 0, \text{ } 1.2 \text{ if } = 1, \text{ } 1.0 \text{ otherwise} \end{aligned}$$

Every multiplier is one the team can justify to a sales-rep audience: outlets without coolers are most responsive to any intervention, dense POI areas amplify response, dense competition dampens it.

Result

LKR **4,999,912.87** allocated across **8,416 of 8,588** outlets; 172 outlets with zero uplift receive zero allocation. No outlet hits the 5,000 LKR cap. The 24-month sensitivity sweep at six elasticity values is published to `outputs/audit/budget_sensitivity.csv` for downstream tuning.

Channel decomposition

Each outlet's allocated `Trade_Spend_LKR` is decomposed into Discount / Merchandising / Promotional using outlet-attribute heuristics (no-cooler outlets overweight merchandising for visibility; high-competition outlets overweight discount + merchandising to defend share; low-spatial-demand outlets overweight promotional to drive trial; default leans merchandising). Province totals:

Channel	LKR	Share
Discount	1,744,750	34.9 %
Merchandising	2,089,088	41.8 %
Promotional	1,166,075	23.3 %
Total	4,999,913	100 %

Sums match allocated to the rupee (rounding drift absorbed in merchandising).

Why we do not claim a single litres figure

The dataset has no record of past promotional campaigns and their volume responses, so β (litres per response-unit) is uncalibrated. Rather than commit to a single guess, the team publishes the expected uplift across plausible β values (e.g. $\beta = 0.025$ implies ≈ 1.04 million L of incremental volume; $\beta = 0.05$ implies ≈ 2.09 million L). A live deployment would calibrate β from a pilot.

Business-decision artifacts (downstream of the LP)

The same κ_i family also feeds five business modules: **Cooler-deployment ROI** (Top-100 by 24-month NPV, total LKR 31.65 M net of LKR 5 M capex at 3.33-month median payback), **outlet action cards** (top-3 prescriptive interventions per outlet), **dormancy-risk early warning** (XGBoost classifier on the 2,707 historical lapse cases, 5-fold CV AUC 0.879), **distributor scorecard** (10-distributor composite z-rank), and **HDBSCAN sales territories** (91 sub-province clusters for route planning).

e. GenAI Transparency Log

Tools and usage policy

Two AI tools were used: an AI coding assistant (chat-based code generation and refactor) and Google Gemini 2.5 Flash, the latter only inside the web application as the XAI narrative renderer. The team chose every methodology, every cohort definition, every constraint rule, every blend weight, every business module, the validation strategy, and the limitations narrative. The AI's role was implementation acceleration on team-specified designs, not design itself. Every output was reviewed against the raw data, the rubric, or domain knowledge before it landed in the repository.

Aggregate audit

Full log at docs/genai_log.md (48 entries with phase, prompt category, AI output, team action, validation step).

Outcome	Count
Accepted as-is	24
Accepted with caveat / verification / documented limitation	17
Rejected and replaced	7
Total entries	48

Where AI accelerated the team most

- Reusable DQ check library boilerplate (six parameterised functions)
- Overpass API fetcher + BallTree haversine spatial join

- Censored-regression literature references and likelihood derivations
- XGBoost surrogate + TreeExplainer SHAP scaffold
- Closed-form water-filling Lagrangian
- FastAPI + Next.js boilerplate
- Gemini API wrapper structure

Where the team critically corrected AI

Selected examples (full list in docs/genai_log.md):

- **Holiday primary key:** the assistant initially suggested Date alone, which would have falsely flagged 98.85 % of holiday rows as duplicates. The team switched to (Date, Holiday_Type) and the duplicate rate dropped to a sensible 56.73 % representing genuine legacy-SFA repeats.
- **Outlet_Type typo count:** an initial counter over-reported corrections (5,958 vs the true 985) because it counted rows whose value happened to match a canonical entry even when no change occurred. Fixed before any narrative was written.
- **Sanity floor:** initial suggestion was `max()` of own history. A single outlier month from a data-entry typo would have inflated the floor for an entire outlet. Replaced with Q95 of own history.
- **Bakry typo missed:** the first normalisation map covered Grocry, Eatry, Pharmcy but missed Bakry (395 outlets — the single largest typo bucket). Caught on a second EDA pass and added.
- **Lat/Lon swap:** a naive `lat/lon ∈ Sri Lanka bbox` rule would have quarantined 240 outlets. The team designed and implemented the swap-recovery heuristic that reclaims 200 of them.
- **Cooler-count counterfactual sign:** the first surrogate run produced negative "add a cooler" deltas for some outlets — a model artefact rather than a domain truth. The team added monotonic constraints (Cooler_Count: +1, competitor_density_norm: -1, has_no_cooler: -1, replenishment_friction: -1) so the counterfactual directions match domain intuition.
- **Gemini output truncation:** the first integration returned ~60 chars of narrative because Gemini 2.5 Flash thinking tokens consume the output budget. The team raised `max_output_tokens` to 4,096 and added a fallback that reads from `response.candidates[ ]` when `.text` errors out.
- **Markdown rendering:** the AI narrative card was displaying raw `**bold**` and `* bullet` markers. The team added `react-markdown + remark-gfm` with explicit Tailwind component mappings.
- **HDBSCAN defaults:** at default parameters HDBSCAN produced four macro-clusters mirroring the provinces — too coarse for sales territory planning. The team tuned `min_cluster_size = 60` with leaf cluster selection and added a "reassign noise to nearest centroid" step, yielding 91 useful sub-province territories.
- **Holiday-multiplier asymmetry:** a uniform 3.6 % under-prediction was traced to one side of the multiplier using `nunique(Date)` while the other used `groupby.size()`. The team standardised both to `nunique(Date)`.

Explainable-AI design (web application)

Each outlet drill-down page in the web app offers an "Explain this outlet" button. When clicked, the FastAPI backend assembles a structured payload — predicted score, top-5 positive and top-5 negative SHAP drivers with signs, the two counterfactual deltas (add a cooler, remove competition), the top-3 ranked recommended actions with rationales, and a compact KPI block — and forwards it to Gemini 2.5 Flash with a deterministic-style system prompt that forbids inventing numbers, forbids statistical jargon, and prescribes a two-paragraph plus three-bullet output shape. Temperature is set to 0.4 so the same inputs

reliably produce similar narratives. The model is positioned as a translator of the structured factual substrate, not as an inference engine; the raw historical volume is never sent.

A live request on a high-potential outlet returns $\approx 1,170$ characters of usable business copy in a single round trip. The frontend renders the response through react-markdown + remark-gfm with explicit Tailwind mappings for paragraphs, bullets, emphasis, and headings, so the narrative appears as proper rich text inside the card rather than as raw markdown.

Most effective prompts

The full audit trail of 48 prompts lives in docs/genai_log.md. Five of them were disproportionately load-bearing for the submission:

1. **Schema survey before code** — *"Survey actual data schemas (column names, row counts, sample rows) for all 5 provided CSVs plus the codebook.xlsx so the plan is built on facts not assumptions."* Caught the Grocry typo, the codebook's reference to a missing Product_Name column, and the monthly (not daily) transaction granularity before any code was written. 2. **Methodology family check** — *"Is supervised ML training appropriate when the problem statement says we are not given a target variable (y)?"* Surfaced the censored-regression family (Tobin 1958, Buchinsky 1998) and informed the team's choice to skip supervised ML entirely. 3. **Constraint rule #4 design** — *"Constraint detection: add a rule that flags Cooler_Count = 0 outlets operating below own_q90 / 3 — suppressed but still selling."* This rule catches the 35 % of outlets that have no cooler and pulled the constraint-detection rate from 37 % to 43.6 %, finally above the team's target band. 4. **Cooler ROI grounding** — *"Per-outlet cooler deployment business case using LKR 50,000 unit cost, 12 % gross margin, 12 % annual cost of capital, 24-month NPV horizon, with monotonic constraints so the counterfactual cannot reward removing a cooler."* Produced the Top-100 list (LKR 31.65 M net 24-month value) that anchors the pitch deck. 5. **XAI system prompt** — *"Write a deterministic-style system prompt that forbids the model from inventing numbers, forbids statistical vocabulary (SHAP, regression, p-value, censored), prescribes a two-paragraph plus three-bullet output shape, and references only the structured payload."* Gave us the Gemini wrapper that reliably produces business-readable narratives across all 20,000 outlets.

Honest limitations surfaced

- β elasticity in the budget LP is uncalibrated — sensitivity sweep published instead of a single number.
- Counterfactuals are model-predicted, not empirically validated (no install-date records in the dataset).
- Cooler replenishment cycle modelled via its *effect* (stockout_flag_sum / active_months), not the delivery process.
- Channel split heuristics from MMM conventions, not data-fitted.
- OSM POI coverage is patchier in rural districts; within-province tier normalisation mitigates this.

— Team DataX